

Chaturvedula Sandilya

sadhilyachaturvedula@gmail.com • [GitHub](#) • [LinkedIN](#) • [\(813 \)-853-4279](#)

EXPERIENCE

Sonline LLC, Georgia, Atlanta, United States

May 2025 - August 2025

MLOps Intern – Python, AWS, Docker, SQL, Federated Learning

- Built and deployed intelligent chatbots on **AWS** using **Botpress** and **SQL** to automate user queries while ensuring scalability and high availability.
- Configured **CI/CD** pipelines and containerized chatbot services with **Docker** to enable reproducible deployments and streamline monitoring in **EC2** environments.
- Integrated **federated learning** into chatbot workflows to enable decentralized model updates without exposing raw user data.
- Applied **differential privacy** techniques during model training to protect sensitive data and align with privacy-preserving AI principles.

Collabera Digital, Bengaluru, India

July 2023 - August 2024

Market Analyst – Power Bi, SEO, SQL, Python, Data Analysis.

- Automated customer feedback analysis by creating data pipelines, reducing manual effort by **30%** and accelerating insight generation.
- Built a **predictive model in Python** with **93%** accuracy to forecast customer behavior, improving marketing campaign effectiveness.
- Analyzed **over 50,000** customer reviews using **Python** and **NLP**, identifying key trends and enhancing customer satisfaction by **18%**.
- Created reusable **Python scripts** to automate repetitive tasks, saving time and boosting team productivity.

PROJECTS

- **Federated Pneumonia Detection** | PyTorch, Flower, Opacus, Docker, ResNet18, Differential Privacy

Developed a federated learning system for pneumonia detection using PyTorch, Flower, and Docker, simulating decentralized training across hospital nodes without sharing raw patient data. Integrated Differential Privacy with Opacus to enforce **HIPAA/GDPR** compliance by injecting calibrated noise into gradients and tracking the privacy budget ($\epsilon \approx 3.0$). Achieved **87%** model accuracy while preserving data security and demonstrating the feasibility of privacy-preserving AI in healthcare.

- **Audio-less Video to Speech Conversion** | AV-HuBERT, PyTorch, TensorFlow, Computer Vision

Built an AI system to convert silent videos into speech, achieving **95.88% accuracy** using AV-HuBERT and computer vision. Optimized processing time by **40%** through self-supervised learning. Research published at **2024 ICIACS**.

- **Homophobia and Transphobia Detection** | NLP, IndicBERT, LaBSE, TensorFlow, Hugging Face

Built a deep learning model to detect homophobic and transphobic content in Dravidian YouTube comments, achieving **94%** accuracy and processing **100,000+** comments for **40%** faster moderation.

TECHNICAL SKILLS

Programming Languages: Python, R, C++, SQL, JS React

Libraries & Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Designing, Implementing, Training, Fine-tuning Machine Learning

Machine Learning & AI: Text classification, Preprocessing, feature engineering, and transforming raw data for machine learning Models, Neural Networks, Federated Learning

DevOps & Cloud: Azure, Google Cloud Platform, Docker, AWS

EDUCATION

MUMA College of Business, University of South Florida, Tampa

August 2024 – Expected May 2026

Master of Science in Artificial Intelligence and Business Analytics, GPA – 3.7 / 4.0

Symbiosis International University (SICSR), Pune, India

May 2020 – June 2023

Bachelor of Business Administration in Information Technology, GPA – 3.6 / 4.0